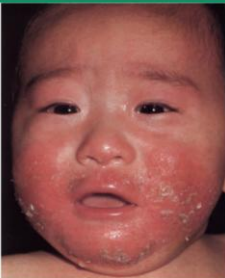
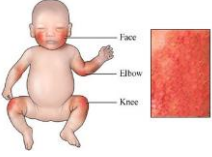
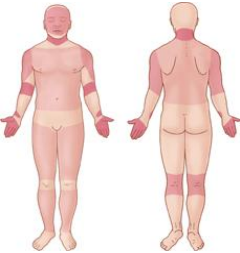
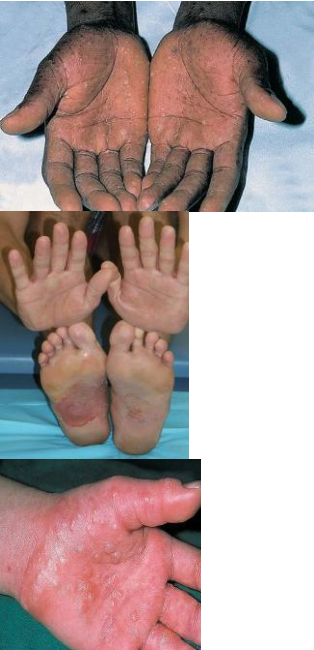


Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Atopic dermatitis	Acute, subacute, or chronic relapsing skin disorder Infantile: Pruritic, red, scaling, crusting lesions White (blanching) dermatographism Atopic dermatitis: infantile  Confluent erythema, microvesiculation, scaling, and crusting on the face, with similar involvement (to a lesser degree) of the trunk and arms. The facial involvement is more severe due to easier access to scratching; the baby is squeezing his breast skin to relieve the intense pruritus. Reproduced with permission from: Fitzpatrick, TB, Johnson, RA, Wolff, et al (Eds). Color Atlas and Synopsis of Clinical Dermatology, 3rd ed, McGraw-Hill, New York, 1997. Copyright © McGraw-Hill. Atopic dermatitis: infantile  Confluent erythema, microvesiculation, papules, crust, and scale of a young Asian infant; the shoulders are relatively spared, being protected from scratching by clothing. Reproduced with permission from: Fitzpatrick, TB, Johnson, RA, Wolff, et al (Eds). Color Atlas and Synopsis of Clinical Dermatology, 3rd ed, McGraw-Hill, New York, 1997. Copyright © McGraw-Hill.	Infantile  Note: Diaper areas usually spared	PREVALENCE : 10-20% of population AGE At ONSET: 85% under 5 85% (age <1 60%)	• Appears to involve genetic defect in proteins supporting epidermal barrier • Impaired epidermal barrier function • Resultant increased water loss (xerosis = dry skin) • Enhanced penetration of potential allergens • Allergen penetrates epidermis, binds to specific IgE on Langerhan's cells • Antigen then presented to T lymphocytes • Predominance of type-2 helper T cells (Th2) • Immune dysregulation • IgE mediated (Type I hypersensitivity?)	Labs/Imaging: Clinical diagnosis (refer if uncertain)...rule out (DDx): • Contact dermatitis • Seborrheic dermatitis • Psoriasis • Drug reactions • Scabies Medications: • Anti-pruritics (topical & systemic) (antihistamines) • Topical glucocorticoids ▫ Possibility of atrophy limits use on face ▫ Very rarely may need to use systemic steroids • Topical and/or systemic antibiotics/antivirals for secondary infections • New topical nonsteroidal anti-inflammatory agents ▫ Do not cause atrophy ▫ Calcineurin inhibitors tacrolimus and pimecrolimus • UVA-UVB phototherapy Patient Education: • Stop the itch cycle ▫ Avoid scratching • Emollients • Limited use of mild soap • Wet dressings • Spontaneous remission in >40% of childhood cases • 30-50% develop asthma and/or allergic rhinitis

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment (Cont.)
Atopic dermatitis (Cont.)	Childhood/adults: Lichenification, hyperpigmentation Atopic dermatitis  Hyperpigmented, slightly scaly patches and lichenified plaque are present in the popliteal fossae of this patient with atopic dermatitis. Reproduced with permission from: www.visualdx.com. Copyright Logica Images, Inc. Adult atopic dermatitis  Chronic atopic dermatitis with lichenification (skin thickening and enhancement of skin markings) of the knee flexures in a 22-year-old woman with atopic dermatitis. Copyright © Monroe Standish, RN, Dermatologist: http://monrodermatitis.org.	Childhood/Adult 	See previous page	See previous page	See previous page

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Asteatotic eczema AKA xerotic eczema, “winter itch”	Dry, scaly skin resulting in erythema, cracks, and fissuring 	Torso and extremities	AGE: Usually elderly TEMPORAL: Winter months	Very dry skin	Labs/Imaging: Clinical diagnosis Medications: Same as for atopic dermatitis
Nummular eczema	“Coin-shaped” (round-to-oval) lesions Often start as papules, which then coalesce into plaques with scale 	Trunk and lower extremities (Head spared)	AGE: 50 to 60 years	• Cause unknown • Spontaneous onset not usually related to identifiable inciting event	Labs/Imaging: Clinical diagnosis; RULE OUT TINEA CORPORIS Medications: Same as for atopic dermatitis

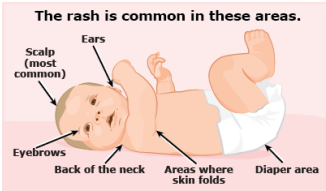
Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Dyshidrotic eczema	- Recurrent or chronic relapsing forms of vesicular palmoplantar dermatitis - Mild form appears as tiny tapioca vesicles, then red, scaly, weeping and pruritic vesicles which may rupture to form enlarging collarettes of scale - Severe form occurs as large bullae on the palms and feet - Burning pain, pruritis 	Sides of the fingers, palms and soles	AGE: young adults GENDER: Male=female	Etiology: Unknown Predisposing factors: - History of atopic dermatitis - Exposure to contact irritants and contact allergens - Smoking Triggers: Stress may be a factor for some	Labs/Imaging: Clinical diagnosis; RULE OUT TINEA CORPORIS, CONTACT DERMATITIS Medications/Procedures: - Topical corticosteroids are first line treatment - Topical calcineurin inhibitors - Systemic corticosteroids for severe disease - Phototherapy Patient Education: Stress reduction if relevant

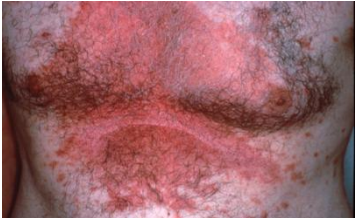
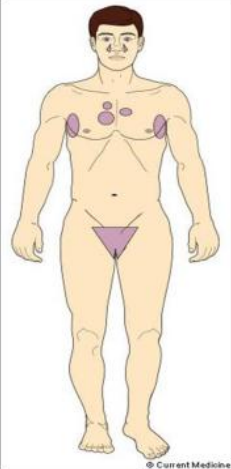
Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
IRRITANT Contact Dermatitis Stinging, smarting →itching Sharp margins, strictly confined to site of exposure Rapid evolution Dependent on concentration of agent and state of skin barrier; occurs only above threshold level May occur in practically everyone	•Erythema with a dull, nonglistening surface •Then vesiculation (or blister formation) •Then erosion, crusting then shedding of crusts and scaling then healing •Chronic: Hyperkeratosis, scaling, fissures In chemical burns: Erythema, then necrosis then shedding of necrotic tissue then ulceration, then healing  Acute, topical cream for back pain  Acute, industrial solvent   Chronic: Top—home use of turpentine to clean brushes after painting Bottom—occupational cement exposure	•Confined to the area of exposure THUS: • Always sharply margined • Never spreads •HANDS the most common location	•Can occur in anyone (having atopic dermatitis increases likelihood) •Occurrence and severity depend on degree of exposure to offending agent (strength and amount) •Most cases caused by chronic cumulative exposure to one or more irritants •Most common form of occupational skin disease (but need not be occupational)	• Exposure of the skin to chemical or other physical agents that are capable of irritating the skin, acutely or chronically • Severe irritants cause toxic reactions even after a short exposure <u>Most common irritant/toxic agents</u> • Soaps • Acids and alkalis • Industrial solvents • Plants	<u>Labs/Imaging:</u> Clinical dx—may need patch test to R/O allergic contact dermatitis, palmar psoriasis if chronic <u>Medications:</u> • Topical & systemic glucocorticoids • Lubricating/protective creams • Oral or topical antibiotics for cracks and fissuring/resistant cases/secondary infection <u>Patient Education:</u> • If contact does occur, wash with water or weak neutralizing solution • Wet dressings • Healing usually occurs within 2 weeks of removal of noxious stimuli • Prevention is key • Identify and remove the etiologic agent • Avoid irritant or caustic chemical(s) by wearing protective clothing (i.e., goggles, shields, gloves) • Barrier creams

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
ALLERGIC Contact Dermatitis Itching → pain Papules may be present Initial sharp margins but may spread (tiny papules) and may become generalized Relatively independent of amount applied Occurs only in the sensitized	• Eczematous (papules, vesicles, pruritic) dermatitis • Acute: Eruption starts 48h to DAYS after initial contact • With phytoallergic (poison ivy), exposure sites may not be apparent • Appearance: Well-demarcated erythema and edema with nonumbilicated vesicles, and/or papules, proceeds to erosions, crusting, scaling • In severe reactions, bullae, confluent erosions exuding serum, and crusts • Local symptoms: Intense pruritus, stinging and pain if severe • Constitutional symptoms in severe cases • Subacute: Plaques of mild erythema showing small, dry scales • Chronic: Small, firm, rounded or flat-topped papules ("satellite" = evidence of spread), plaques of lichenification, scaling, excoriations, erythema, and pigmentation repeated exposures lead to a crescendo reaction	• Site of the eruption is confined to site of exposure • May spread	• Occurs only in sensitized individuals • NOT dependent on degree of exposure to offending agent (even small amount can cause extreme reaction)	• Systemic disease defined by hapten-specific T cell-mediated inflammation • Due to reexposure to a substance to which the individual is sensitized • Classic, delayed, cell-mediated hypersensitivity reaction	Labs/Imaging: Clinical diagnosis Medications: • Topical steroids may be sufficient if limited reaction • Widespread disease may require systemic corticosteroids such as a burst and taper of prednisone Patient Education: • Eliminate offending agent • Usually when agent is removed dermatitis will run its course in 2-3 weeks

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
ALLERGIC Contact Dermatitis (Cont.)	 Agent: Eosin in lip cream  Agent: Occupational chromate exposure  Latex allergy  Nickel allergy	See previous page	See previous page	Ten most common agents: • Nickel sulfate • Neomycin sulfate • Balsam of Peru • Fragrance mix • Thimerosal • Sodium gold thiosulfate • Formaldehyde • Quaternium-15 • Bacitracin • Cobalt chloride	See previous page

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Subset: Allergic phytodermatitis	Acute, very pruritic, eczematous dermatitis, often in a linear arrangement  Poison ivy  Poison ivy	Site of exposure	Occurs in <u>sensitized</u> individuals	Occurs after exposure to a wide variety of plant allergens	See previous pages
Attention! <u>Phytophotodermatitis</u> can mimic allergic reaction (more like IRRITANT contact dermatitis than ALLERGIC)	Hyperpigmentation common	Sun-exposed areas of body	Can occur in anyone	• Photosensitivity reaction • Specific plant-derived chemical on the skin causes skin reaction when exposed to sunlight • Common agents: • lime/citrus juice • celery • parsley	

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Seborrheic dermatitis Synonyms: "Cradle cap" (infants), pityriasis sicca (dandruff)	  	Occurring in regions where the sebaceous glands are most active: • Face and scalp • Presternal area • In body folds Mild scalp SD causes flaking, i.e., dandruff 	Age of Onset: • Infancy (within the first months) • Puberty • Adults: Most between 20 and 50 years or older Gender: Males > females Associated conditions: Increased prevalence in patients with HIV or Parkinson's disease	• Very common chronic skin condition characterized by redness and scaling • <i>Malassezia furfur</i> is said to play a role in the pathogenesis	Labs/Imaging: Clinical diagnosis Medications: • Topical corticosteroids and/or antifungals (ketokonazole) • If severe = systemic antifungals or oral retinoids • Shampoos containing ketoconazole or antiproliferative agents: salicylic acid, selenium, tar, sulfur or zinc Patient Education: • Behavioral modification (stop scratching) • Mineral oil and gentle brushing for cradle cap • Early treatment of flares is encouraged

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Seborrheic dermatitis (Cont.)	•Orange-red or gray-white skin •Often with "greasy" or white dry scaling macules, papules of varying size (5-20 mm), or patches •Sharply marginated •Nummular, polycyclic, and even annular on the trunk   Severe seborrheic dermatitis: Before treatment  After isotretinoin	 •Behind external ear = Sticky crusts and fissures •Scalp = diffuse, marked scaling ("dandruff") •Face & trunk = scattered, discrete lesions	See previous page	See previous page	See previous page

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Stasis Dermatitis		Lower extremities primarily	Common in patients with chronic venous insufficiency 6.2% of patients over age 65	- Acute: inflamed, weeping plaques, vesicles, crusting  - Chronic: hyperpigmentation, scaling, lipodermatosclerosis 	Treatment: <u>Symptomatic:</u> topical steroids, emollients <u>Must treat underlying cause (venous insufficiency):</u> - Leg elevation - Weight loss - Compression therapy - Venous ablation

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Urticaria AKA “hives” <u>CLINICAL</u> <u>ALERT:</u> Can be harbinger of life-threatening anaphylaxis so look for associated history and physical findings!!!	• Pruritic, edematous, raised, pink plaques without scale that move and change daily • Annular lesions with central clearing and white halos can also occur   	Anywhere on body		• Results from histamine released from tissue mast cells or circulating basophils • May be IgE-mediated (usually acute), chemical-induced urticaria (non-IgE-mediated), urticarial vasculitis, other forms • Acute (<6 weeks), or Chronic (>6 weeks) • Possible triggers: • Food - fruits, nuts, soy, wheat, fish, etc. • Drugs – antibiotics, etc. • Latex – condoms, gloves, catheters, etc. • Insect bites and stings • Infection- CMV, hepatitis C, chronic dental infection, H. pylori, etc. • Internal disease - Autoimmune diseases, cancer, liver disease, etc. • Inhalants – dust, animal dander, etc. • Stress – emotional or physical • Sun • Also from physical triggers: stroking (dermatographism), pressure (pressure urticaria), or cold (cold urticaria) • Undetermined (up to 60%; as much as 80% in chronic urticaria)	<u>Labs/Imaging:</u> Clinical diagnosis, a thorough history is key! • If food likely: Patch test, RAST, food diary <u>Medications:</u> Potentially life-threatening situations: • If associated with angioedema – give IM epinephrine • If bronchospasm – give nebulized albuterol Other symptomatic treatment: • Antihistamines are first-line therapy • Usually IM or PO diphenhydramine • Consider glucocorticoids for severe cases • They stabilize mast cells and prevent histamine release <u>Patient Education:</u> • Avoid causative agent if identified • If potentially life-threatening reaction: • Medical Alert bracelet • Epinephrine pen

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Angioedema <u>CLINICAL ALERT:</u> Potentially life-threatening	• Edema of entire parts of the body • May be preceded by urticarial • Mortality can occur secondary to respiratory tract involvement 	Commonly eyes, lips, genitalia, hands and feet		• Deep, subcutaneous, submucosal edema • Mast cell mediated • Pathogenesis is similar to urticaria but manifests in deeper mucosal tissue • Types • Hereditary angioedema, types I, II and III • Acquired angioedema • Drug-induced angioedema (ACEI, ARBs) • Urticaria-associated angioedema • Idiopathic angioedema • Can be episodic and self-limiting • Often recurrent	Potentially life-threatening • Same as urticaria, but early airway management is key

Disorder	Clinical Presentation	Affected Areas	Epidemiology	Etiology/Pathogenesis and Triggers	Plan/Appropriate Treatment
Stasis Dermatitis	Erythematous, scaling, and eczematous patches or plaques on chronically edematous legs Acute forms may present with severely inflamed, weeping plaques, vesiculation, and crusting Chronic forms are characterized by hyperpigmentation, due to dermal hemosiderin deposition, scaling, and potential development of lipodermatosclerosis (fat degeneration and fibrosis) 	Common in lower extremities 	Affects 6.2% of patients over the age of 65	Due to chronic venous insufficiency	Management involves the treatment of the underlying chronic venous insufficiency • Leg elevation • Weight reduction • Compression therapy • Venous ablation therapy Symptomatic treatment of skin dryness, pruritus and prevention of ulceration Topical corticosteroids for acute erythema, pruritus, vesiculation, and oozing